

Comparing Three Numerical Methods for Planar Two-Link Arm Motion

Single Shooting (RK4) · Trapezoidal Collocation · Hermite–Simpson Collocation

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Move the arm from pose A to pose B with least torque

Find joint trajectories $q(t)$ and torques $\tau(t)$, $t \in [0, T]$, that solve

$$\min_{q, \tau} \int_0^T \left(\|\tau\|^2 + w \|\dot{q}\|^2 \right) dt, \quad w = 10^{-3}$$

subject to $M(q) \ddot{q} + C(q, \dot{q}) \dot{q} + g(q) = \tau$ (physics, at every instant)

$$q(0) = q_A, \quad q(T) = q_B, \quad \dot{q}(0) = \dot{q}(T) = 0$$

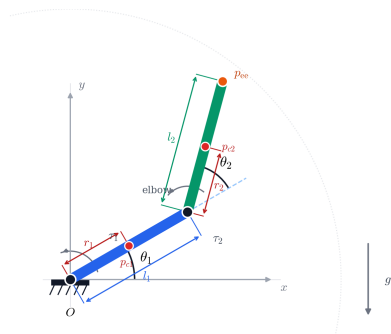
$$|\tau| \leq \tau_{\max}, \quad |\dot{q}| \leq \dot{q}_{\max}$$

There is no closed-form solution: the dynamics must hold at infinitely many instants. **Direct transcription** keeps only $N+1$ time points and turns this into a finite nonlinear program, solved here with SciPy's SLSQP.

Takeaway: we compare three transcriptions on the *same* model, cost and tasks, so every difference comes from the method.

1 · PROBLEM

The two-link arm is a 4-state, 2-input nonlinear system



State $x = [\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2]^\top$, control $u = [\tau_1, \tau_2]^\top$, $q = [\theta_1, \theta_2]^\top$.

$$M_{11} = m_1 r_1^2 + I_1 + m_2 (l_1^2 + r_2^2 + 2l_1 r_2 \cos \theta_2) + I_2$$

$$M_{12} = M_{21} = m_2 (r_2^2 + l_1 r_2 \cos \theta_2) + I_2, \quad M_{22} = m_2 r_2^2 + I_2$$

$$C\dot{q} = \begin{bmatrix} -h(2\dot{\theta}_1\dot{\theta}_2 + \dot{\theta}_2^2) \\ h\dot{\theta}_1^2 \end{bmatrix}, \quad h = m_2 l_1 r_2 \sin \theta_2$$

$$g(q) = g_0 \begin{bmatrix} (m_1 r_1 + m_2 l_1) \cos \theta_1 + m_2 r_2 \cos(\theta_1 + \theta_2) \\ m_2 r_2 \cos(\theta_1 + \theta_2) \end{bmatrix}$$

Every method calls the same function:

$$\dot{x} = f(x, u) = \begin{bmatrix} \dot{q} \\ M(q)^{-1}(u - C\dot{q} - g) \end{bmatrix}$$

Uniform rods: $l_i = 1 \text{ m}$, $m_i = 1 \text{ kg}$,
 $r_i = 0.5 \text{ m}$, $I_i = \frac{1}{12} \text{ kg}\cdot\text{m}^2$, $g_0 = 9.81 \text{ m/s}^2$.

Method 1 — Single shooting: optimise torques, simulate states

Unknowns are only the torques $u_0, \dots, u_{N-1}$ ($2N$ variables), each held constant over its interval. The states come from one classical RK4 step per interval:

$$x_{k+1} = x_k + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$k_1 = f(x_k, u_k), \quad k_2 = f(x_k + \frac{h}{2}k_1, u_k), \quad k_3 = f(x_k + \frac{h}{2}k_2, u_k), \quad k_4 = f(x_k + hk_3, u_k)$$

The only constraint is the end state: $x_N(u_0, \dots, u_{N-1}) = x_B$.

Each step is $\mathcal{O}(h^4)$ accurate, so accuracy is not the weakness — **conditioning** is: x_N chains N steps together, so a tiny change in u_0 is amplified through every later step.

Takeaway: few variables, but a long and sensitive chain from the first torque to the final state.

Method 2 — Trapezoidal collocation: states become unknowns

Unknowns are the state *and* torque at every node, $z = [x_0, u_0, \dots, x_N, u_N]$, i.e. $6(N+1)$ variables. Physics is enforced as a *defect* between neighbouring nodes (trapezoid rule):

$$x_{k+1} - x_k = \frac{h}{2} \left(f(x_k, u_k) + f(x_{k+1}, u_{k+1}) \right), \quad k = 0, \dots, N-1$$

The cost uses the same rule on the running cost $\ell = \|u\|^2 + w\|\dot{q}\|^2$:

$$J = \sum_{k=0}^{N-1} \frac{h}{2} (\ell_k + \ell_{k+1})$$

Accuracy is $\mathcal{O}(h^2)$ globally. Each defect touches only two neighbouring nodes, so an error at one time cannot spread along the whole trajectory, and the constraint Jacobian is banded and sparse.

Takeaway: more variables than shooting, but every equation is local and well conditioned.

Method 3 — Hermite–Simpson: add a midpoint per interval

Separated form: the midpoint state x_c and control u_c are independent unknowns, giving $6(2N+1) = 12N + 6$ variables. Two constraints per interval:

$$\text{cubic midpoint: } x_c - \frac{1}{2}(x_k + x_{k+1}) - \frac{h}{8}(f_k - f_{k+1}) = 0$$

$$\text{Simpson defect: } x_{k+1} - x_k - \frac{h}{6}(f_k + 4f(x_c, u_c) + f_{k+1}) = 0$$

with $f_k = f(x_k, u_k)$. The cost uses Simpson's weights:

$$J = \sum_{k=0}^{N-1} \frac{h}{6}(\ell_k + 4\ell_c + \ell_{k+1})$$

The state is a cubic spline and the control is piecewise quadratic; the error shrinks as $\mathcal{O}(h^4)$.

Takeaway: roughly twice the unknowns per interval in exchange for fourth-order accuracy.

The three methods at a glance

	Shooting (RK4)	Trapezoidal	Hermite–Simpson
Unknowns	$2N$	$6(N+1)$	$12N + 6$
Physics enforced	by simulation	between 2 neighbours	2 neighbours + midpoint
Accuracy	$\mathcal{O}(h^4)$ per step	$\mathcal{O}(h^2)$	$\mathcal{O}(h^4)$
Conditioning	poor on long horizons	good	good
Control between nodes	held constant	linear	quadratic

Takeaway: the last row matters later: it decides how a torque schedule must be replayed to be judged fairly.

Four test scenarios of increasing difficulty

Scenario	Motion	T	$\tau_{\max}$	What makes it hard
Baseline	$[0, 0] \rightarrow [\pi/2, 0]$	1.0 s	30 N·m	reference case
Obstacle	$[-\pi/4, 0] \rightarrow [\pi/4, 0]$	1.2 s	40 N·m	circle $r = 0.35$ m at (1.2, 0)
Swing-up	$[-\pi/2, 0] \rightarrow [\pi/2, 0]$	2.0 s	6 N·m	too weak to lift (stall ≈ 14.7 N·m)
Fast maneuver	$[-\pi/3, \pi/4] \rightarrow [\pi/2, -\pi/3]$	0.45 s	80 N·m	strong velocity coupling

All solvers: SLSQP, at most 400 iterations, tolerance 10^{-5} , straight-line initial guess with gravity-holding torque.

Physical check: optimal torques replayed in SciPy's adaptive DOP853 integrator (tolerance 10^{-8}).

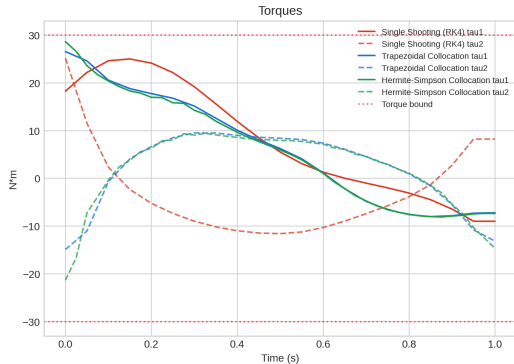
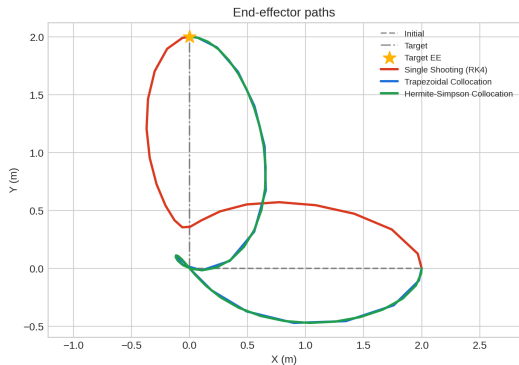
Baseline: Trapezoidal fastest, Hermite–Simpson least effort

Metric ($N = 20, T = 1.0 \text{ s}$)	Shooting	Trapezoidal	Hermite–Simpson
Solve time (s)	6.37	3.14	12.26
Optimizer iterations	53	39	54
Torque effort $\int \ \tau\ ^2 dt$	282.9	223.4	217.8
Peak torque (N·m)	25.1	26.5	28.6

All three report success. Hermite–Simpson’s cubic description of the motion finds a trajectory that needs about **23% less effort than shooting**; Trapezoidal solves in half of shooting’s time with the fewest iterations.

Takeaway: collocation wins on both speed and effort on the reference task.

Collocation methods agree; shooting finds a different motion



Takeaway: the two collocation curves overlap almost exactly; shooting settles in another, costlier local optimum.

Replay drift measures the torque convention, not the method

First reading: shooting says it hits the goal, but the DOP853 replay lands **1.77 m** away. The replay interpolated torques *linearly*; shooting assumed them *held constant* (zero-order hold).

Scenario	Method	Linear replay	Zero-order-hold replay
Baseline	Shooting	1.771 m	0.234 m
	Trapezoidal	0.071 m	2.973 m
	Hermite–Simpson	0.280 m	1.793 m
Obstacle	Shooting	1.317 m	0.034 m
	Trapezoidal	0.267 m	2.995 m
	Hermite–Simpson	0.405 m	3.594 m

Final end-effector drift. Bold = the convention that solver actually assumes.

Takeaway: every method is accurate under its own convention; we withdrew the “shooting is unphysical” claim.

Where shooting really fails: long, unstable motions

Swing-up ($T = 2$ s, $\tau_{\max} = 6$ N·m): the arm must swing back first to build momentum.

- Shooting **does not converge**: 400 iterations, ≈ 1420 s, constraint violation 1.4×10^{-3} .
- Trapezoidal and Hermite–Simpson both succeed: cost 60.55 and 60.51, zero terminal error.

Fast maneuver ($N = 25$, $T = 0.45$ s):

Metric	Shooting	Trapezoidal	Hermite–Simpson
Solve time (s)	24.4	55.2	101.1
Torque effort	1336.6	1214.4	1208.4
Peak torque (N·m)	80.0 (at limit)	74.7	77.0
Max replay drift (m)	—	0.020	0.023

Takeaway: Hermite–Simpson’s higher order did *not* beat Trapezoidal at $N = 25$: theory is a hypothesis, not a result.

Five changes to the Trapezoidal solver, same equations

- S1 Fix start and end as bounds** (lower = upper) instead of equality constraints: 8 fewer constraints.
- S2 Batch the physics:** evaluate $f(x_k, u_k)$ at all nodes in one array operation, with the closed-form 2×2

$$\text{inverse } M^{-1} = \frac{1}{\det M} \begin{bmatrix} M_{22} & -M_{12} \\ -M_{12} & M_{11} \end{bmatrix}.$$

- S3 Sparse Jacobian by graph colouring.** Finite differences cost one evaluation per variable,

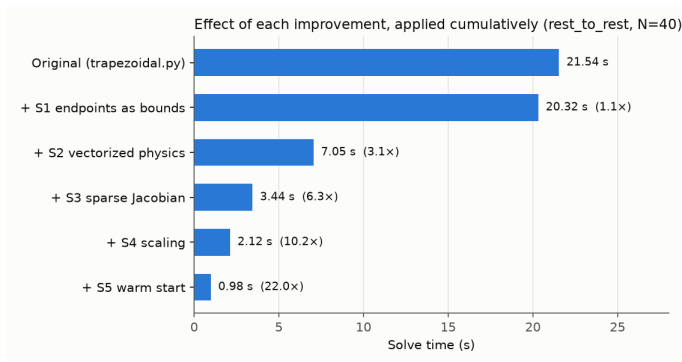
$$\frac{\partial F}{\partial z_c} \approx \frac{F(z + \epsilon e_c) - F(z)}{\epsilon},$$

but variables that never share a defect can be perturbed together: always **12 groups** instead of $6(N+1)$ ($= 246$ at $N = 40$).

- S4 Rescale:** $y = z/s$ from $\dot{q}_{\max}$, $\tau_{\max}$, and the cost by $h \tau_{\max}^2$ (scaling variables alone made it *worse*: 115 iterations vs 46).

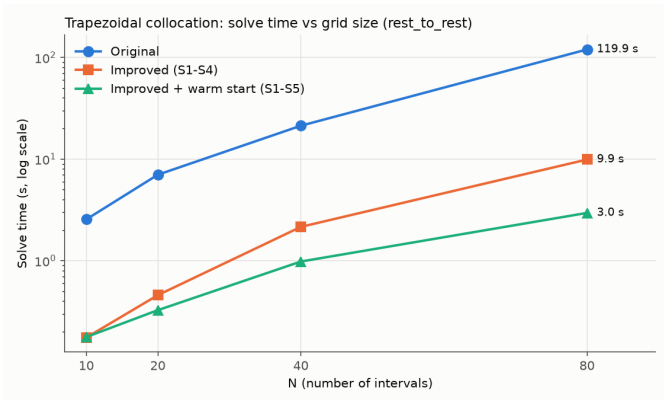
- S5 Warm start:** solve on a coarse grid, interpolate up, repeat ($10 \rightarrow 20 \rightarrow 40 \rightarrow 80$).

Each change removes a different bottleneck



Takeaway: $21.54\text{ s} \rightarrow 0.98\text{ s}$ ($22\times$) at $N = 40$ with the same cost; batching the physics is the biggest single jump.

The speed-up grows with grid size



N	Original	All five	Speed-up
10	2.55 s	0.18 s	14.4×
20	7.01 s	0.33 s	21.5×
40	21.30 s	0.98 s	21.7×
80	119.87 s	2.95 s	40.6×

Takeaway: the colouring stays at 12 groups for any N , so larger problems gain more.

The speed-up holds on every scenario

Scenario ($N = 30$)	Original	Improved	Speed-up	Answer
Baseline	14.53 s	0.61 s	23.8×	same
Obstacle	28.78 s	3.74 s	7.7×	29% cheaper
Fast maneuver	34.63 s	0.81 s	42.9×	same
Under-torqued	12.73 s	0.42 s	30.3×	same
Reversal	93.54 s	1.66 s	56.4×	same
Swing-up	19.23 s	0.83 s	23.1×	same

Obstacle gains least (its constraint is not vectorised) and lands in a different valid local minimum (239.2 vs 335.3).

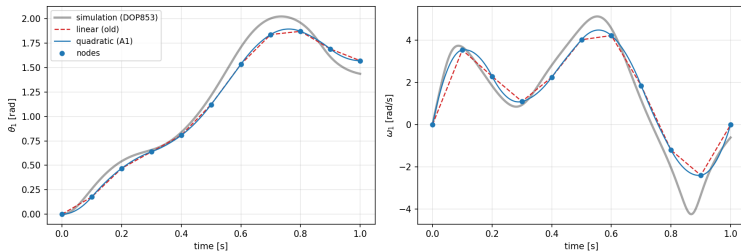
Takeaway: the full configuration is both the fastest and the most reproducible default.

Between nodes, trapezoidal collocation implies a parabola

If $\dot{x} = f$ is linear over an interval, integrating once gives, for $0 \leq \sigma \leq h$,

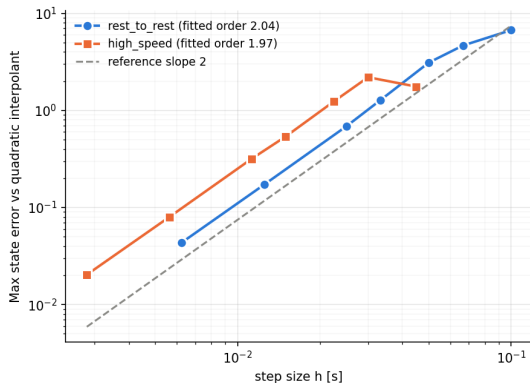
$$x(t_k + \sigma) = x_k + f_k \sigma + \frac{\sigma^2}{2h} (f_{k+1} - f_k), \quad u(t_k + \sigma) = u_k + \frac{\sigma}{h} (u_{k+1} - u_k)$$

Trapezoidal state between nodes (rest_to_rest, N = 10)



Takeaway: the parabola matches the true motion; straight lines between nodes do not.

Measured convergence confirms second order



Theory: $N \times 2 \Rightarrow h \div 2 \Rightarrow \text{error} \div 2^2 = \div 4$.

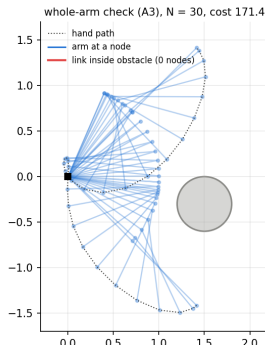
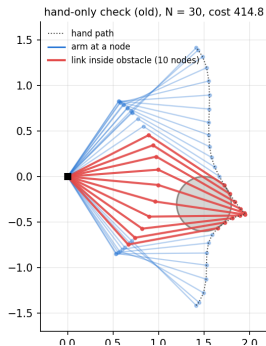
Max state error	all N	$N \geq 20$
Baseline	1.90	2.04
Fast maneuver	1.76	1.97

From $N = 40 \rightarrow 80 \rightarrow 160$ the error falls $4.0\times$ and $3.9\times$ per doubling. Constraint violation stays below 9×10^{-13} .

Takeaway: the $\mathcal{O}(h^2)$ claim is now a measurement, not an assumption.

Checking only the hand let the links pass through the obstacle

New check, per link segment $[a, b]$: $s^* = \text{clamp}\left(\frac{(c-a) \cdot (b-a)}{\|b-a\|^2}, 0, 1\right)$, $\|a + s^*(b-a) - c\| \geq r + 5 \text{ mm}$.



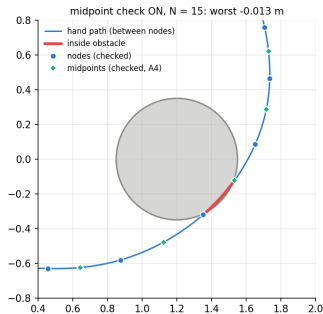
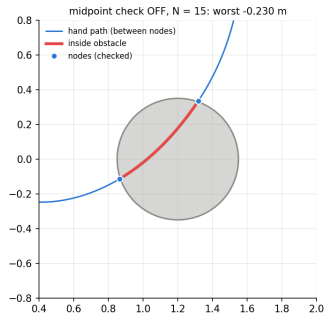
N	Check	Cost	Min clear.
20	hand	417.2	-0.300
20	arm	172.5	+0.230
30	hand	414.8	-0.300
30	arm	171.4	+0.230
40	hand	394.7	-0.299
40	arm	171.0	+0.230

Takeaway: “success” held for the constraint we wrote, not for the arm; the whole-arm check folds the elbow instead.

Checking between nodes stops corner cutting

Also enforce the constraint at every interval midpoint, from the parabola at $\sigma = h/2$:

$$x_{\text{mid}} = x_k + \frac{h}{8} (3f_k + f_{k+1})$$



Worst hand penetration (m)

N	Midpts off	Midpts on
8	-0.302	-0.333
10	-0.261	-0.156
15	-0.230	-0.013
20	-0.126	-0.003
30	-0.015	+0.002

Takeaway: collision-free from $N = 30$ at +24% effort; halving the gap is no guarantee ($N = 8$ still fails).

What we learned

1. **Collocation is the practical default.** Trapezoidal is fastest; Hermite–Simpson uses least effort; both solve the swing-up that shooting cannot.
2. **Solver engineering matters as much as the method.** Five changes give 22–41 $\times$ on the baseline and up to 56 $\times$ across scenarios, with the same answer.
3. **Optimizer success $\neq$ physical correctness.** The obstacle runs satisfied every constraint while both links went through the obstacle.
4. **Check what a metric really measures.** Shooting's 1.77 m drift was mostly our replay convention; under its own it is 0.234 m.
5. **Theory is a hypothesis.** $\mathcal{O}(h^2)$ was confirmed (2.04); Hermite–Simpson's expected edge at $N = 25$ was not.

Limitations and next steps

- Whole-arm and midpoint checks exist only in the improved Trapezoidal solver; move them into all three.
- Replay each method under its own control convention everywhere (done only for baseline and obstacle).
- Run a coarse sweep ($N = 8\text{--}15$) to expose Hermite–Simpson's fourth-order advantage.
- Replace the midpoint check with a certified bound over each interval.
- Vectorise the obstacle constraint; try adaptive time steps and a 6–7 joint arm, as in the base paper.

Cardona-Ortiz and Arechavaleta, AMCS 35(4) 577–589, 2025 · Kelly, SIAM Review 59(4) 849–904, 2017 · Tedrake, Underactuated Robotics ch. 10, 2024 · Betts, SIAM 2010

Thank you

Questions and discussion

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